

# HW 8

1.  $0.3 R_{old} C_L = R_{new} 2 C_L$

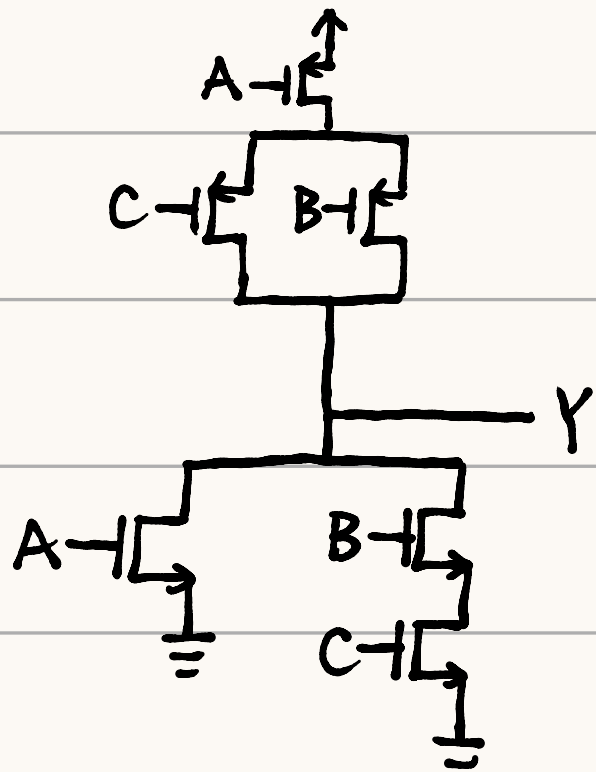
$R_{new} = 0.15 R_{old}$

$\frac{1}{0.15} = 6.67$

$\therefore (\frac{W}{L})_{n, new} = 1.5 \times 6.67 = 10$

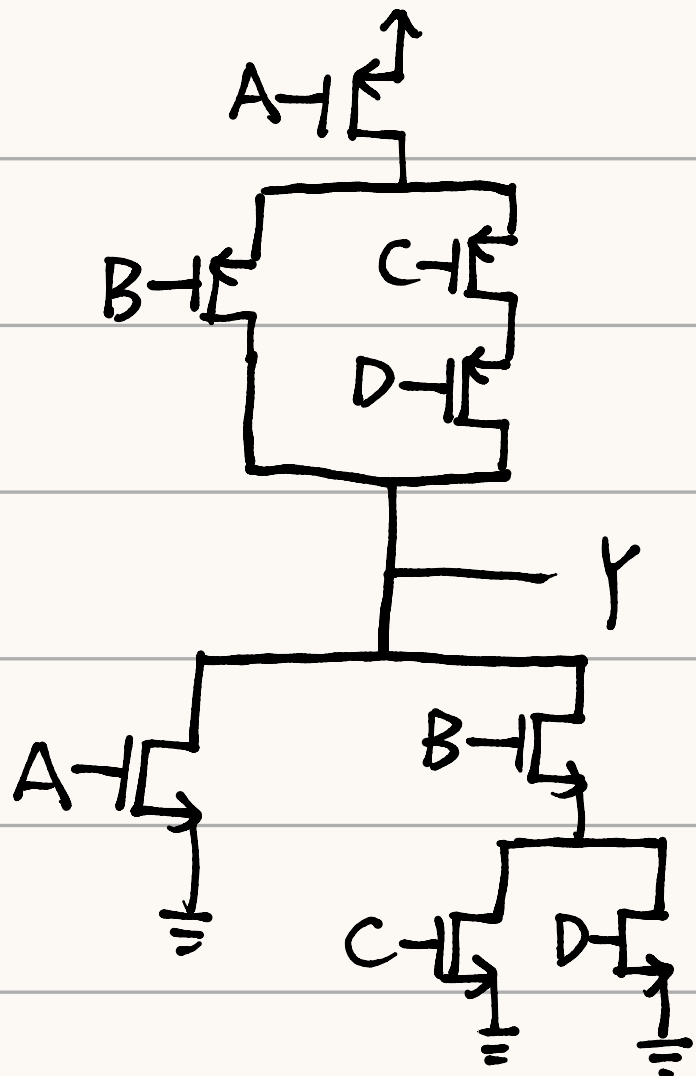
$(\frac{W}{L})_{p, new} = 10 \times 6.67 = 66.7$

2.  $Y = \overline{A+BC}$      $\bar{Y} = A+BC$      $Y = \bar{A}(\bar{B}+\bar{C})$



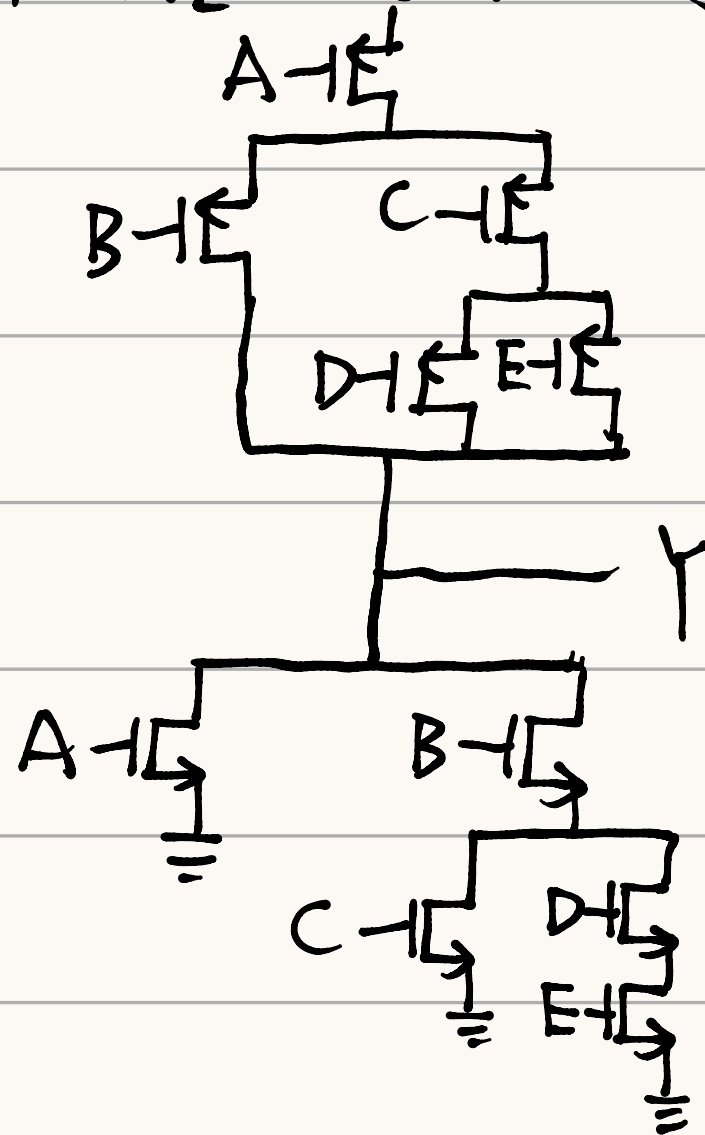
3.  $Y = \overline{A+B(C+D)}$      $\bar{Y} = A+B(C+D)$

$Y = A' (B' + c'D')$



4.  $Y = \overline{A+B(C+DE)}$      $\bar{Y} = A+B(C+DE)$

$Y = \bar{A}[\bar{B}+\bar{C}(\bar{D}+\bar{E})]$



5.  $\bar{Y} = F + (C+E)[G+(B+D)A]$

$Y = \overline{F + (C+E)[G+(B+D)A]}$